PIECEWISE FUNCTIONS · CONTINUATION

From Pieces to Pictures

Solving, building and graphing piecewise functions

In Problems 1–4 we...

- read piecewise definitions
- plugged values into the right piece
- checked continuity at joining points

In this part we will...

- build a clear problem-solving recipe
- see what these functions are *for*
- graph them properly — and solve 3 more

What's inside

Method (slide 2) · Purpose (slide 3) · Graphing guide (slides 4–5) · Three new problems with full solutions (slides 6–11) · Wrap-up (slide 12)

THEORY · METHOD

How to solve piecewise problems

A reliable five-step recipe that works on every problem in this deck

1. **Read the function as a table.**
   Each row is one piece — a formula *f(x)* together with an interval that says when the formula applies (its domain).

2. **Locate *x* in an interval.**
   For the input you are given, ask: which piece does it belong to? At a boundary, check whether the inequality is strict (<, >) or inclusive (≤, ≥).

3. **Pick exactly one piece.**
   Substitute the value into the matching formula. Never average two pieces and never use both at once.

4. **Compute and label.**
   Evaluate carefully. Write the result with units (dollars, hours, °C…) so the meaning is clear.

5. **Sanity-check at the joins.**
   Plug a boundary value into both pieces. If the values agree, the function is continuous; if not, mark the jump.

Pitfalls to avoid

- using the wrong piece because you skimmed the inequality
- forgetting that ≤ includes the endpoint, < does not
- plugging into both pieces and adding the answers
- dropping units or signs in the second piece

Mantra · *"One x, one piece, one answer."*

Going backwards · if the *output* is given (e.g. "find x so that f(x) = 20"), find which piece's range contains 20, then solve that piece's equation for x.

THEORY · PURPOSE

Why we build piecewise functions

Real life rarely follows one rule from start to finish

A **piecewise function** is a single function whose rule changes when the input crosses certain boundaries. We use them whenever a real-world quantity follows *different rules in different ranges*, but we still want one formula that captures the whole behaviour.

💸 Money & pricing

- Tax brackets, electricity tariffs, shipping costs
- "First N units cheap, then more expensive"
- Phone & data plans (Problem 1) and overtime pay

🚚 Motion & physics

- An object that accelerates, then cruises, then stops
- Heating water from ice → liquid → steam
- Toll fees that change at a fixed distance

📋 Rules & thresholds

- Grade boundaries: 0–49 fail, 50–69 pass, 70+ merit
- Working hours: regular pay vs overtime
- Speed limits that change between zones

The big idea

A piecewise function takes *messy reality* — one rule here, a different rule there — and packages it into a single mathematical object we can evaluate, graph, optimise and reason about. That is why engineers, economists, scientists and game programmers all use them daily.

Question to keep in mind · for any quantity you study, ask *"does the rule ever change?"* The moment the answer is yes — at a threshold, a deadline, a boundary — you have a piecewise function on your hands.

GUIDE · PART 1

Choosing the dimensions of your graph

Six decisions to make before the pencil touches paper

1. **Identify the variables.**
   What is on the horizontal axis (input x) and on the vertical axis (output $f(x)$)? Always include the units.

2. **Find the domain you need.**
   Look at every piece. The domain is the union of all sub-intervals — for example $x \geq 0$, or $0 \leq x \leq 12$.

3. **Find the range.**
   Evaluate every piece at its endpoints (and any vertex of a parabola). The smallest output gives the lower bound, the largest gives the upper bound.

4. **Add a comfort margin.**
   Extend each axis ~10% past the data so the graph doesn't feel cramped against the edge.

5. **Pick a sensible scale.**
   Choose tick spacing in round numbers (1, 2, 5, 10, 20, 50, 100…). The two axes can use different scales — match the data.

6. **Mark the joins.**
   Plot a filled dot ● where the piece *includes* the endpoint, and an open dot ○ where it *excludes* it.

Quick example

Decisions for the parking-cost graph (slide 6)

Decision	Choice
x-axis	t (hours)
y-axis	C(t) (dollars)
x range	0 → 24
y range	0 → 22
x ticks	every 2 h
y ticks	every $5

Tip · If a piece is a *line*, you only need its two endpoints to draw it. If it is a *parabola* or a curve, plot at least 5 points so the shape is honest.

GUIDE · PART 2

Drawing the graph, piece by piece

A worked walk-through using a small two-piece function

Our function

$$f(x) = \begin{cases} x + 2 & \text{if } x < 1 \\ -x + 4 & \text{if } x \geq 1 \end{cases}$$

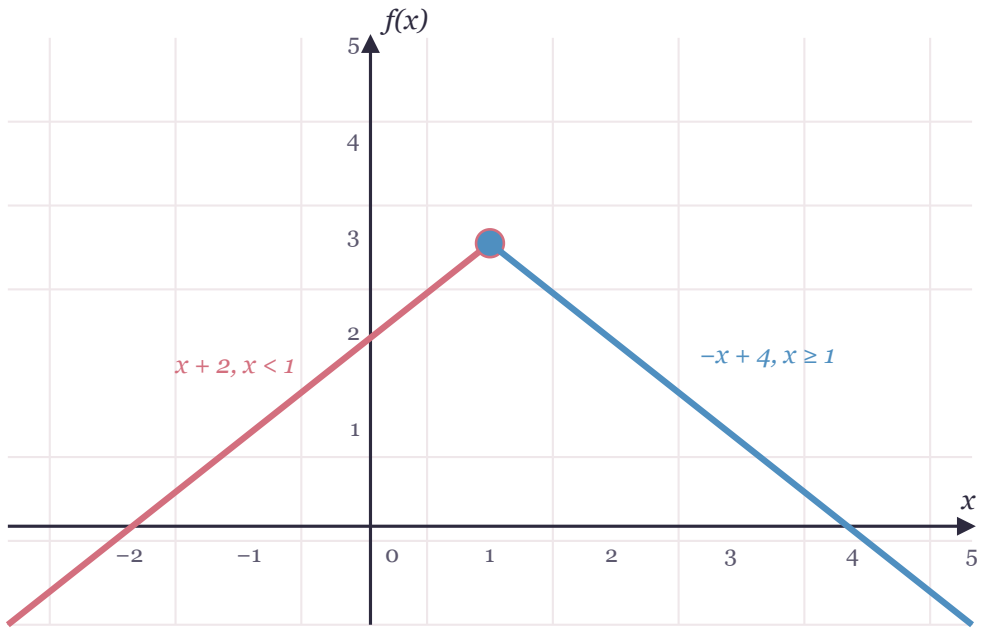
1. **Draw the axes.**
   Mark equal ticks on both. Label them with the variable names.

2. **Plot piece 1.**
   For $x < 1$, plot two points such as $(-2, 0)$ and $(0, 2)$. Continue the line up to $x = 1$ and end with an
   open
   dot at $(1, 3)$.

3. **Plot piece 2.**
   For $x \geq 1$, plot $(1, 3)$ with a
   filled
   dot, then $(4, 0)$. Draw the segment between them.

4. **Read the join.**
   Both pieces meet at $(1, 3)$ — the function is *continuous*. If they hadn't, you would mark the gap with both an open and a filled dot.

Piecewise example · f(x)



Piece 1 (open at join) — Piece 2 (filled at join)

Open vs filled dots · ≤ or ≥ ⇒ filled, < or > ⇒ open. At a continuity point one piece supplies the filled dot and the open from the other piece sits underneath — visually they overlap, which is fine.

PROBLEM 5

Parking garage rates

A flat starting fee, then a per-hour charge with a daily cap

A parking garage in town charges drivers as follows:

- $**4** flat fee for parking up to and including 1 hour
- $**2** per additional hour after the first hour
- A maximum charge of $**20** once you have parked 9 hours or more (up to 24 h)

Cost function

$$C(t) = \begin{cases} 4 & \text{if } 0 \le t \le 1 \\ 4 + 2(t-1) & \text{if } 1 < t < 9 \\ 20 & \text{if } 9 \le t \le 24 \end{cases}$$

where t is the number of hours parked.

FIND

- (a) Cost for 0.5 hour
- (b) Cost for 1 hour
- (c) Cost for 4 hours
- (d) Cost for 12 hours

Notice · The numbers are chosen so the function is *continuous*: piece 2 evaluated at $t = 9$ gives $4 + 2(8) = 20$, which agrees with piece 3.

SOLUTION 5

Parking garage rates

Pick the matching piece, then evaluate

(a) $t = 0.5$ · since $0 \le 0.5 \le 1$, use the first piece

C(0.5) = $4

(b) $t = 1$ · boundary, top piece (≤)

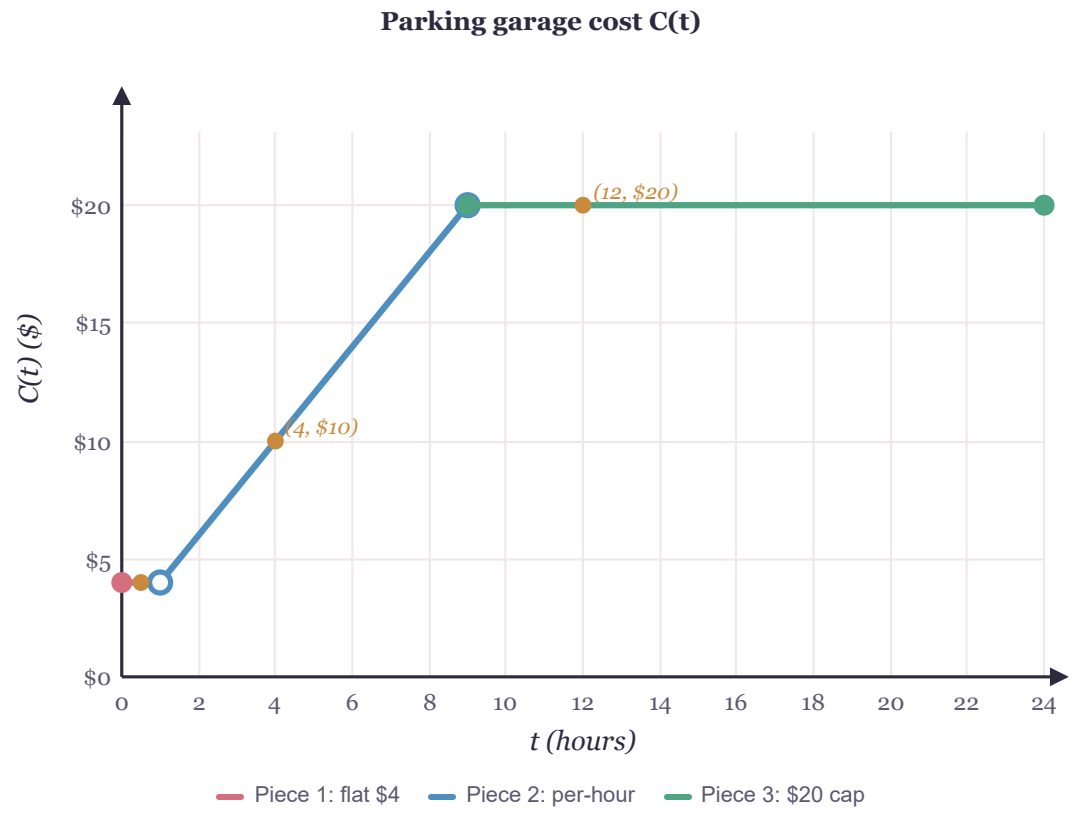
C(1) = $4

(c) $t = 4$ · since $1 < 4 < 9$, use the middle piece

$C(4) = 4 + 2(4 - 1) = 4 + 6 =$ **$10**

(d) $t = 12$ · since $9 \le 12 \le 24$, the daily cap applies

C(12) = $20



The cost grows linearly between hours 1 and 9, then locks at the daily cap. The function is continuous everywhere — no jumps.

PROBLEM 6

Weekly pay with overtime

A standard hourly rate, then time-and-a-half after 40 hours

A worker is paid:

- $**15** per hour for the first 40 hours of the week
- $**22.50** per hour ("time-and-a-half") for any hour beyond 40
- The week is capped at 60 hours

Pay function

$$P(h) = \begin{cases} 15h & \text{if } 0 \le h \le 40 \\ 600 + 22.5(h - 40) & \text{if } 40 < h \le 60 \end{cases}$$

where h is the number of hours worked in the week.

FIND

- (a) Pay for 25 hours
- (b) Pay for 40 hours
- (c) Pay for 50 hours
- (d) How many hours produce a pay of $825?

Going backwards · part (d) is a *reverse* question. First decide which piece has output 825, then solve that piece's equation.

SOLUTION 6

Weekly pay with overtime

Forward ($h \to P$) and reverse ($P \to h$) — both directions matter

(a) $h = 25$ · since $0 \le 25 \le 40$, top piece

$P(25) = 15 \times 25 = \mathbf{\$375}$

(b) $h = 40$ · boundary, top piece (≤)

$P(40) = 15 \times 40 = \mathbf{\$600}$

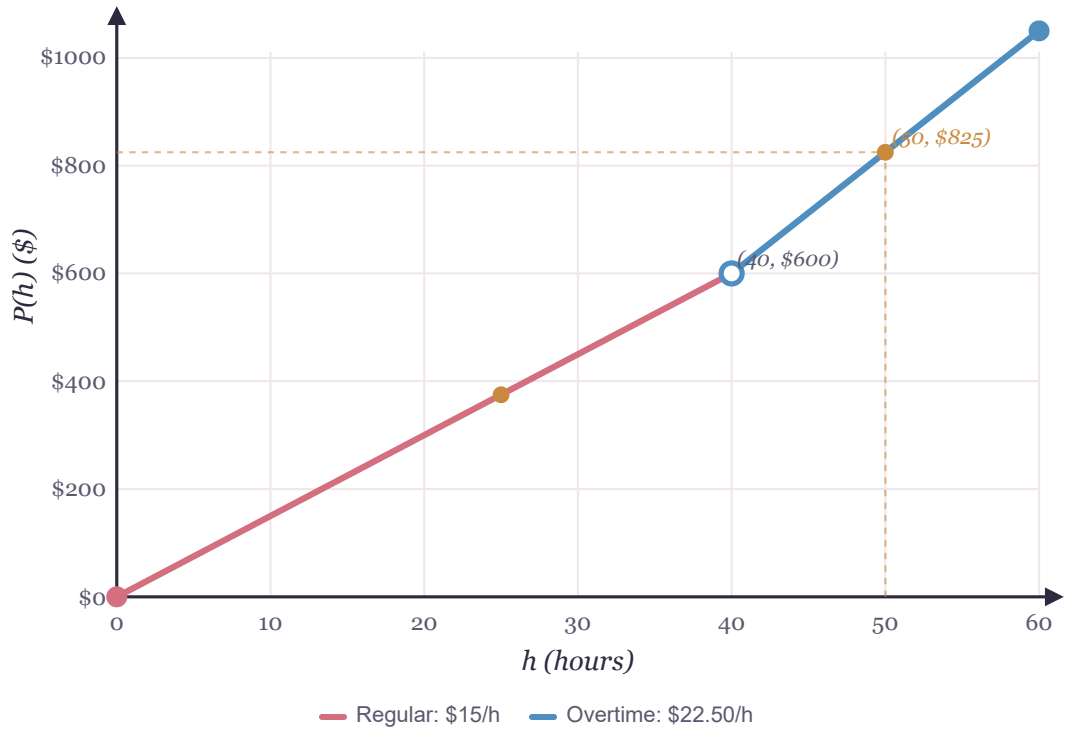
(c) $h = 50$ · since $40 < 50 \le 60$, bottom piece

$P(50) = 600 + 22.5(50 - 40) = 600 + 225 = \mathbf{\$825}$

(d) Pay \$825 > \$600, so we are in the *bottom* piece. Solve:

$825 = 600 + 22.5(h - 40)$
$225 = 22.5(h - 40)$
$h - 40 = 10 \Rightarrow \mathbf{h = 50\ hours}$

Consistent with part (c) — a useful self-check.



The graph is one connected curve — the function is **continuous**. Only the *slope* changes at h = 40, and the steeper slope is the overtime rate.

PROBLEM 7

Find the constant — continuity

*A classic exam favourite: choose **k** so the pieces fit perfectly*

Consider the function

$$f(x) = \begin{cases} 2x + 1 & \text{if } x \le 2 \\ x^2 + k & \text{if } x > 2 \end{cases}$$

- The first piece is a straight line
- The second piece is a parabola, shifted up by k
- For the function to make sense as a single graph, the two pieces must *meet* at $x = 2$

FIND

- (a) The value of **k** that makes f continuous at $x = 2$
- (b) $f(2)$ with that k
- (c) $f(5)$ with that k

Continuity test · the value the left piece gives at the boundary must equal the value the right piece gives there.

SOLUTION 7

Find the constant — continuity

Set the two pieces equal at the join, then solve for k

(a) Force the pieces to agree at $x = 2$

left piece at 2: $2(2) + 1 = 5$
right piece at 2: $(2)^2 + k = 4 + k$
set equal: $4 + k = 5 \Rightarrow$ **k = 1**

k = 1

(b) $f(2)$ with k = 1

Use the top piece (it includes $x = 2$):

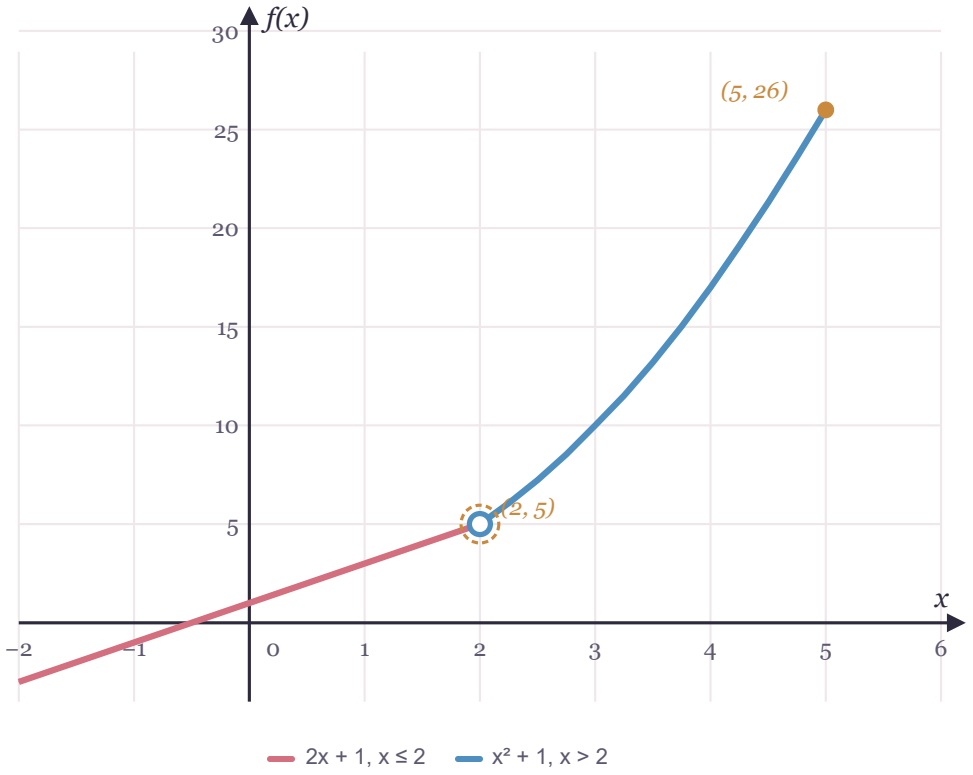
$f(2) = 2(2) + 1 = \mathbf{5}$

(c) $f(5)$ with k = 1

Use the bottom piece (5 > 2):

$f(5) = 5^2 + 1 = 25 + 1 = \mathbf{26}$

Continuous piecewise function (k = 1)



With **k = 1** the pieces meet exactly at *(2, 5)* — no jump. For any other value of k the parabola would land above or below 5 and the graph would tear.

WRAP-UP

What you can now do

Three big skills, ready to take into any new piecewise problem

1 · Solve

- Locate x in the right interval
- Substitute into one piece only
- Verify continuity at every join
- Reverse-solve when the output is given

2 · Build

- Translate "rule changes here" into intervals
- Pick consistent variables and units
- Make sure every input has exactly one output
- Decide if the function should be continuous

3 · Graph

- Choose axes, range, scale and ticks deliberately
- Use filled ● and open ○ dots correctly
- Plot lines from endpoints; curves from samples
- Annotate joins, kinks and jumps clearly

One sentence to remember

"A piecewise function is one function with several rules — pick the rule by checking the interval first."

End of *Piecewise Functions — Continuation*